

elementsetup

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17:09

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1 Routines used to set up the elements file

\$Id: 3fba2f30a3210e3829fc5e685d348c61a996c042 \$

The `elements_infile` identified in the `degas2.in` file lists all of the elements available to DEGAS 2 for use in constructing the species. As shown in this example, the atomic number, a symbol, the atomic mass (in AMU), and a descriptive name are provided. The program *datasetup* reads this file, translates this information into DEGAS 2 data, and generates the netCDF file with the symbolic name `elementsfile`.

#	Z	Sy	amu	name
	0	0	0	geometry
-1	e		5.4858e-4	electron
0	n		1.008665	neutron
1	H		1.007825	hydrogen
1	D		2.0140	deuterium
1	T		3.01605	tritium
2	He3		3.01603	helium-3
2	He		4.002602	helium
3	Li		6.941	lithium

```
"elementsetup.f" 1 ≡
  @m FILE 'elementsetup.web'
```

The unnamed module.

```
"elementsetup.f" 1.1 ≡
  ⟨Functions and Subroutines 1.2⟩
```

Read in data from `elements_infile`.

⟨ Functions and Subroutines 1.2 ⟩ ≡

```

subroutine read_elements
  implicit none_f77
  el_common    // Common
  rf_common
  implicit none_f90
  st_decls
  integer length, p, b, e    // Local
  character*LINELEN line
  character*FILELEN tempfile    // local

  ⟨ Memory allocation interface 0 ⟩

  el_num = 0
  element_version = 'unknown'
  tempfile = filenames_array_elements_infile
  assert(tempfile ≠ char_undef)
  open(unit = diskin, file = tempfile, form = 'formatted', status = 'old')
  assert(read_string(diskin, line, length))
  assert(length ≤ len(line))
  element_version = line(: length)

loop1: continue
  if (¬read_string(diskin, line, length))
    goto eof
    el_num = el_num + 1
    var_realloca(element_sy)
    var_realloca(element_name)
    var_realloca(element_m)
    var_realloca(element_z)
    length = parse_string(line(: length))
    p = 0
    assert(next_token(line, b, e, p))
    el_z(el_num) = read_integer(line(b : e))
    assert(next_token(line, b, e, p))
    el_sy(el_num) = read_text(line(b : e))
    assert(next_token(line, b, e, p))
    el_m(el_num) = read_real(line(b : e))
    assert(next_token(line, b, e, p))
    el_name(el_num) = read_text(line(b : e))
    goto loop1
eof: continue
  close(unit = diskin)
  var_reallocb(element_sy)
  var_reallocb(element_name)
  var_reallocb(element_m)
  var_reallocb(element_z)
  return
end

```

See also section 1.3.

This code is used in section 1.1.

Write out data into netcdf file **elements.nc**

⟨Functions and Subroutines 1.2⟩ +=

```
subroutine nc_write_elements
  implicit_none_f77
  el_common    // Common
  rf_common
  implicit_none_f90
  nc_decls    // Local
  st_decls
  integer fileid
  character*FILELEN tempfile    //local

  el_ncdecl
  character*LINELEN description, program_version

  program_version = 'DEGAS_2_Git_commit:_'$Format:%H$, _ref_names:_'$Format:%d$'

  tempfile = filenames_array_elementsfile
  assert(tempfile ≠ char_undef)
  fileid = nccreate(tempfile, NC_CLOBBER, nc_stat)

  description = 'Data_for_elements_in_degass_2'
  call ncattputc(fileid, NC_GLOBAL, 'description', NC_CHAR, string_length(description),
    description, nc_stat)

  call ncattputc(fileid, NC_GLOBAL, 'data_version', NC_CHAR, string_length(element_version),
    element_version, nc_stat)

  call ncattputc(fileid, NC_GLOBAL, 'program_version', NC_CHAR,
    string_length(program_version), program_version, nc_stat)

  el_ncdef(fileid)
  call ncendef(fileid, nc_stat)
  el_ncwrite(fileid)

  call ncclose(fileid, nc_stat)

  return
end
```

2 References

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assert: 1.2, 1.3.

b: 1.2.

char_undef: 1.2, 1.3.

datasetup: 1.

description: 1.3.

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var_realloca: 1.2.

var_reallocb: 1.2.

⟨ Functions and Subroutines 1.2, 1.3 ⟩ Used in section 1.1.

⟨ Memory allocation interface 0 ⟩ Used in section 1.2.

COMMAND LINE: "fweave -f -i! -W[-ybs15000 -ykw800 -ytw40000 -j -n/
/Users/dstotler/degas2/src/elementsetup.web".

WEB FILE: "/Users/dstotler/degas2/src/elementsetup.web".

CHANGE FILE: (none).

GLOBAL LANGUAGE: FORTRAN.